P-release kinetics as a predictor for P-availability in Swiss cropping systems

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2026-08-19

Abstract

Traditional static soil tests (STPs) for phosphorus (P) often fail to predict agronomic outcomes because they do not account for the kinetic nature of P supply to plant roots. This study investigated whether P desorption kinetic parameters could serve as more effective predictors. Using soils from the long-term Swiss agricultural experiment (STYCS), a sequential extraction method was refined and modeled with a non-linear approach to derive the maximum equilibrium P intensity (P_{desorb}) and a rate constant (k). The predictive power of these kinetic parameters was compared against standard Swiss STPs (P_{CO_2} and P_{AAE10}) using linear mixed-effects models. For predicting site-normalized yield, STPs were superior, while for national-normalized yield and P-uptake, both methods performed poorly due to dominant pedoclimatic factors. However, the kinetic model demonstrated higher accuracy in predicting the long-term P-Balance, with P_{desorb} alone explaining 57% of the variance, whereas STP models showed no predictive power. We conclude that while traditional STPs remain adequate for short-term within-field fertility, kinetic parameters serve as a more robust functional proxy for assessing the long-term P status and sustainability of agricultural soils.

1 Introduction

1.1 The Agronomic Dilemma of Phosphorus

Phosphorus (P) is an essential macronutrient for all life, yet its management in agricultural systems presents a significant challenge. In the soil solution, P exists predominantly as highly reactive orthophosphate anions (HPO_4^{2-} or $H_2PO_4^-$). These dissolved species are rapidly immobilized—either adsorbed onto the surfaces of clays and metal oxides or precipitated with cations like calcium, iron, and aluminum to form minerals of low solubility (Brady & Weil, 2016; Sposito, 2008). Consequently, the total soil P stock is vast, but the minute fraction dissolved in the soil solution—the only form directly available to plant roots—is highly transient.

To manage this complex nutrient, European agricultural systems rely heavily on empirical soil tests to generate fertilizer recommendations. As detailed by Jordan-Meille et al. (2012) and later expanded by Steinfurth et al. (2022), these systems generally follow a three-step procedure: soil testing, calibration, and recommendation. Central to this process is the "Class Concept" (Klassenkonzept), where raw soil test values are calibrated into discrete P-fertility categories (e.g., Very Low, Target, Excessive). In many European countries, including Switzerland and France, these empirical thresholds are further modified by secondary soil parameters such as clay content and pH.

However, despite these sophisticated calibration matrices, inter-laboratory comparisons reveal massive contradictions, with recommended fertilizer doses for the exact same soil varying by up to 300% across Europe (Jordan-Meille et al., 2012). This fundamental instability arises because standard chemical extractants largely ignore the mechanistic processes governing plant P uptake.

1.2 Conceptual Frameworks vs. Operational Definitions

To conceptualize plant P availability, soil science relies on three fundamental thermodynamic and mass-balance parameters (Frossard et al., 2000): 1. **P-Intensity (I)**: The thermodynamic energy state, or chemical activity, of P in the soil solution at any given moment. 2. **P-Quantity or Capacity (Q)**: The physical mass of solid-phase P that is readily desorbable and can replenish the soil solution.

3. **Phosphorus Buffering Capacity (PBC):** The resistance of the soil to changes in P-Intensity, mathematically represented as the derivative of Quantity with respect to Intensity (dQ/dI).

While these concepts are theoretically robust, measuring them in natural soils is methodologically challenging. Isotopic Exchange Kinetics (IEK) perfectly capture these dynamics by tracing the true exchangeable pool without altering the chemical equilibrium (Fardeau, 1996; Gallet et al., 2003). However, the requirement for radioisotopes renders IEK unsuitable for routine agronomic testing. In practice, standard agronomic soil tests do not measure absolute thermodynamic states; rather, they rely on **operational definitions** that often misrepresent the true pool of available P (Demaria et al., 2005). A soil test value is strictly defined by the specific chemical extractant used, the extraction time, and the soil-to-solution ratio.

Different standard methods target different physicochemical pools of P by employing distinct extraction mechanisms: - **Olsen P** ($NaHCO_3$, pH 8.5): Measures desorbable P. The high concentration of bicarbonate ions outcompetes orthophosphate for adsorption sites on soil surfaces (Olsen et al., 1954). - **P-AAE10 (Ammonium Acetate EDTA, pH 4.65):** A strong extractant that attacks both sorbed and precipitated P forms. EDTA chelates background cations, actively dissolving apatites, while the acetate buffer facilitates desorption (Forschungsanstalt für Agrarökologie und Landbau (FAL), 1996). - **Water Extractions (P_{H_2O} and P_{CO_2}):** While lacking aggressive chelators, distilled or CO_2-saturated water acts as an incredibly strong thermodynamic sink. By plunging the soil into a massively dilute environment, it forcefully drives dissolution and desorption to re-establish equilibrium. Consequently, the actual extracting power of these methods is fiercely modulated by the physical extraction time and, crucially, the soil-to-extractant ratio.

1.3 The Asymmetry of Measurement: Thermodynamics vs. Biology

Relying on operational definitions introduces severe analytical limitations, but a profound epistemological asymmetry exists between the chemical measurements of the soil and the biological measurements of the plant.

On the soil side, thermodynamic measurements can be mechanistically clarified. While empirical tests like P_{AAE10} suffer from clear artifacts (such as the physical saturation of the EDTA chelator in heavy clays), we can rationally address these limitations. We can convert stoichiometric concentrations to corrected chemical activities (γ) using Davies or Pitzer equations, isolate kinetic bottlenecks, and model fundamental isotherms. We have rational, physical control over the soil matrix.

Conversely, biological response variables are inherently messy, abstract, and riddled with physiological ambiguity. Agronomic models attempt to use plant tissue concentrations (C_P) or total Phosphorus Uptake (P_{uptake}) as response variables, but these are not absolute physical truths. They are abstractions derived from indirect measurements of mass and concentration.

Furthermore, these biological metrics routinely collapse in the field due to physiological phenomena like the **Piper-Steenbjerg effect** (carbon dilution) (Hirte et al., 2021). When a severely P-starved plant finally receives Phosphorus, it rapidly generates new biomass. This burst of growth paradoxically *dilutes* the tissue C_P concentration, causing the measured concentration to drop even though the total Phosphorus mass in the plant increased. This immediately forces an arbitrary decision: which specific plant tissue should we optimize to represent the whole organism?

Finally, even if a perfect thermodynamic soil model is constructed, its predictive power in field trials remains constantly threatened by overarching pedoclimatic noise. Much like the Athenians praying to Zeus for rain in the *Meditations* of Marcus Aurelius, agronomic field trials are ultimately subservient to variables beyond human control. Temperature, precipitation, and soil moisture exert effect estimates that vastly overpower subtle chemical variations. Therefore, while we can mathematically anchor our thermodynamic extractions, predicting biological field outcomes requires accepting a fundamentally chaotic and uncertain reality.

1.4 Unifying Quantity, Intensity, and the Kinetic Bottleneck

By pairing an Intensity measurement with a mass extraction (Q, such as P_{AAE10}), we can construct a generalized $Q(I)$ framework. We parameterize this relationship using a Freundlich isotherm ($Q = K_f \cdot I^{1/n}$) not because it yields a "true" universal constant, but because its log-log linearization elegantly isolates the baseline capacity (K_f) and the saturation breakdown (n). This thermodynamic abstraction liberates the analysis from the arbitrary, method-specific bounds of operational extractants (e.g., AAE10 vs. CAL vs. Olsen). It translates disparate empirical measurements into a common, transparent language where the Phosphorus Buffering Capacity (PBC) can be derived as the continuous first derivative ($\frac{dQ}{dI}$) and meaningfully compared across borders.

However, even with a correct thermodynamic understanding of Q and I, plant uptake remains fundamentally limited by time. Standard conceptual models assume instantaneous solid-solution equilibrium. In reality, desorption is a rate-limiting kinetic bottleneck (Flossmann & Richter, 1982; Nye & Tinker, 2000). When a root depletes the rhizosphere, the soil cannot always resupply phosphate fast enough.

To address this, we define a dynamic predictor: the **Initial Activity Flux** (J_0). This flux shifts the focus from static mass to supply rate by multiplying the first-order desorption rate constant (k) by the maximum thermodynamic activity (a_{max}):

$$J_0 = k \cdot a_{max}$$

Table 1 maps this transition from classic, operationally defined static tests to the proposed thermodynamic and kinetic parameters.

Table 1: Conceptual mapping of fundamental phosphorus availability factors to their analytical proxies.

Approach / Framework	Intensity Factor	Quantity (Capacity) Factor	Kinetic Factor
Theoretical Concept (Frossard et al. (2000))	Concentration of P in the soil solution	Pool of solid-phase P that can replenish the solution	Rate of P transfer from the solid to the liquid phase
Traditional / Static Proxy (GRUD)	P_{CO_2} (Raw Water-soluble Concentration)	P_{AAE10} (Chelate-extractable Mass)	*Not Measured*
Dynamic / Kinetic Proxy (This Study)	Raw Intensity (P_{CO_2})	P_{AAE10}	Dynamic Desorption Velocity ($v_0 = k \cdot P_{CO_2}$)

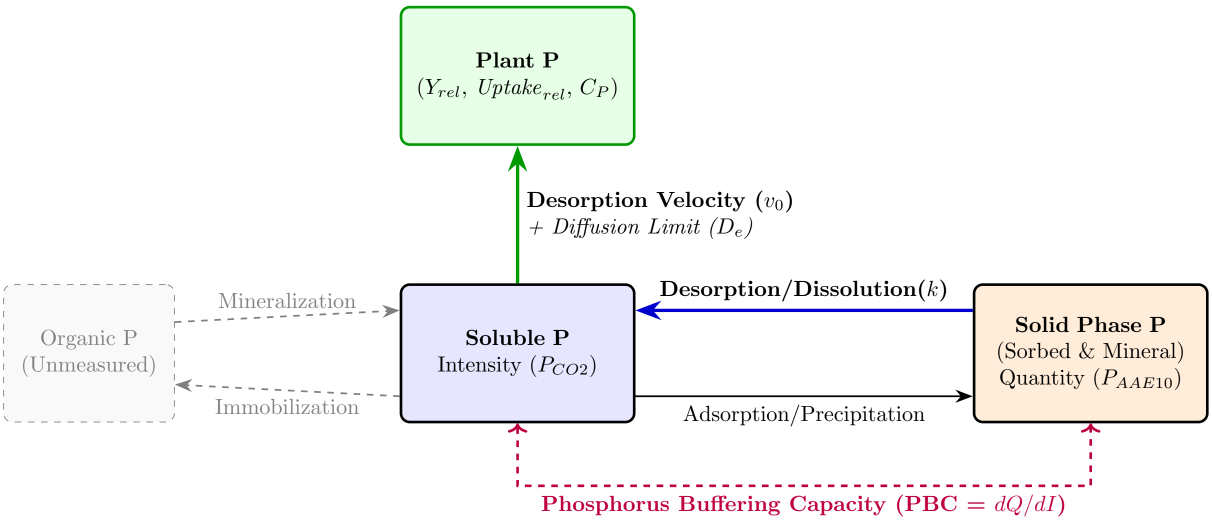

Figure 1: Conceptual model of the rhizosphere phosphorus cycle.

1.5 Thesis Objectives and Analytical Strategy

This thesis hypothesizes that integrating thermodynamic activity and desorption kinetics provides a more comprehensive mechanistic framework for P-management compared to standard empirical testing. Specifically, it directly answers the call by Jordan-Meille et al. (2012) to replace empirical extraction classes with mechanistic sorption-desorption models.

Using soils from the long-term Swiss agricultural experiment STYCS, this study aims to validate this framework through the following core objectives and research questions:

1. **Thermodynamic Framework Formulation:** Establish generalized $Q(I)$ isotherms across highly diverse pedoclimatic sites to abstract the buffering capacity into universally comparable parameters (K_f and n). *Research Question:* Is the Swiss GRUD framework and its analytical proxies (P_{CO_2}, P_{AAE10}) compliant with the broader European class concept presented by Jordan-Meille et al. (2012), and do these empirical classes map meaningfully onto the underlying thermodynamic abstractions of physical desorption kinetics?
2. **Agronomic Prediction:** Statistically evaluate the predictive power of the kinetic Activity Flux (J_0) against relative crop yield and P-uptake. To isolate the chemical P-limitation from overarching climatic noise, agronomic outcomes will be normalized against a historical biological baseline (the historical maximum yield of non-limiting P treatments for specific site-crop combinations). *Hypothesis:* The thermodynamically corrected J_0 will significantly outperform standard static extractions.
3. **Mechanistic Validation:** Propose a modified conceptual framework for radial transport to mathematically illustrate that the desorption rate acts as the primary limiting bottleneck restricting P-availability in the rhizosphere.

2 Materials and Methods

2.1 The Long-Term Phosphorus Fertilization Experiment

The soil samples for this thesis originate from a set of six long-term field trials in Switzerland, established by Agroscope between 1989 and 1992. The primary objective of these experiments was to validate and re-evaluate Swiss phosphorus (P) fertilization guidelines by assessing long-term crop yield responses to varying P inputs across different pedoclimatic conditions. A detailed description of the experimental design and site characteristics can be found in Hirte et al. (2021).

The experiment was set up as a **completely randomized block design** with four field replications at each site. The core of the experiment consists of six fixed-plot treatments representing different P fertilization levels. These treatments were applied annually as triple superphosphate (TSP) before tillage and sowing. The application rates were strictly calibrated as percentages of the site- and crop-specific officially recommended P inputs (the Swiss GRUD norm): 0% (Zero), 33% (Deficit), 67% (Reduced), 100% (Norm), 133% (Elevated), and 167% (Surplus). This rigorous gradient ensures a massive divergence in cumulative P-balances across the 30-year timeframe, forcing distinct pedological responses.

Crucially, all other macronutrients—including Nitrogen (N), Potassium (K), and Magnesium (Mg)—were applied uniformly across all plots to prevent secondary yield limitations. However, it must be noted that these "non-limiting" applications were structurally calibrated to exactly 100% of their respective empirical Swiss GRUD recommendations, derived from the same matrix of operational soil tests and crop removal estimates criticized in this study. Consequently, the assumption that secondary nutrients were strictly non-limiting throughout the 30-year trial relies entirely on the epistemological validity of the empirical GRUD framework itself—a paradox that underscores the urgent need to transition toward mechanistic, site-specific nutrient modeling.

2.1.1 Crop Yield, Phosphorus Uptake, and Balances

While the fundamental experimental design and baseline data collection follow the protocols described in Hirte et al. (2021), the specific processing of the agronomic variables for this study diverges to explicitly isolate the chemical P-limitation.

Crop yields were measured annually at harvest. Total fresh biomass was recorded, and representative subsamples were oven-dried to determine the absolute dry mass yield. Plant tissue phosphorus concentration (C_P) was subsequently quantified from these dried subsamples. The total phosphorus uptake (P_{uptake}) for each plot was calculated stoichiometrically as the product of the dry mass yield and the measured C_P.

To accurately quantify the historical legacy of the different fertilization treatments, the **Cumulative P Balance** (*Cum_P_bal*) was calculated for each individual plot. *Cum_P_bal* is defined as the net sum of all P inputs (from the annual TSP fertilization) minus all P outputs (specifically, the P removed from the field via the harvested crop biomass) accumulated over the entire 30-year duration of the trial. This provides a direct, continuous metric of the anthropogenic P surplus or deficit, which drives the underlying thermodynamic buffering systems.

2.2 Analytical Workflow and Data Integration

To answer the research questions, the methodology is structured into three highly synchronized, modular steps mapping the raw data to the final agronomic predictions:

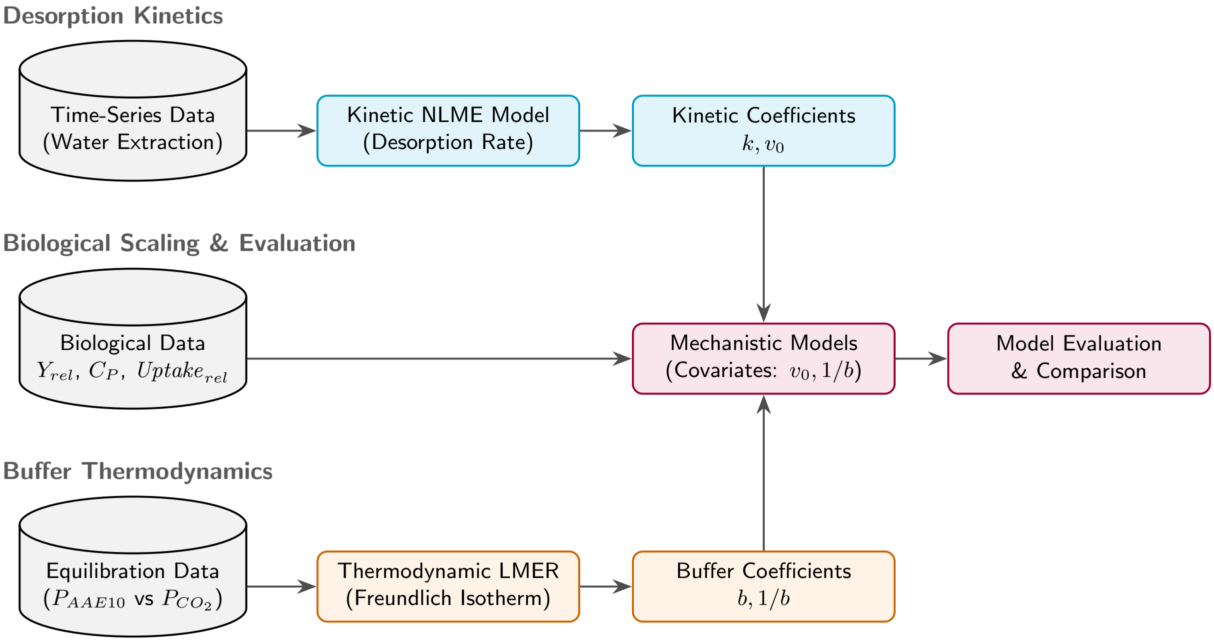

Figure 2: Modular workflow synchronizing data sources, extraction methods, and mathematical models.

2.3 Experimental Sites

The six experimental sites are located in the main crop-growing regions of Switzerland: **Rümlang-Altwi (ALT)**, **Cadenazzo (CAD)**, **Ellighausen (ELL)**, **Grabs (GRA)**, **Oensingen (OEN)**, and **Zurich-Reckenholz (REC)**. The key soil properties are summarized below.

Table 2: Soil characteristics of the six long-term experimental sites. Data adapted from Hirte et al. (2021).

Site	Soil Type (WRB)	Clay (%)	Silt (%)	Sand (%)	Organic C (g/kg)	pH (H2O)
ALT	Calcaric Cambisol	22	30	48	21	7.9
CAD	Eutric Fluvisol	8	52	40	14	6.3
ELL	Eutric Cambisol	33	36	31	23	6.6
GRA	Calcaric Fluvisol	17	49	34	16	8.3
OEN	Gleyic-calc. Cambisol	37	31	32	24	7.1
REC	Eutric Gleysol	39	36	25	27	7.4

Soil samples for this thesis were collected in the year 2022 from the topsoil layer (0-20 cm).

2.4 Phosphorus Desorption Kinetics and Thermodynamics

The laboratory extraction of phosphorus was based on the sequential kinetic principles established by Flossmann & Richter (1982), subsequently modified to capture high-resolution temporal kinetics and corrected to reflect thermodynamic activity.

2.4.1 Adapted Kinetic Protocol for This Study

To capture the desorption process with high temporal resolution while preserving the native soil matrix, a modified water-extraction protocol was employed:

1. **Pre-washing to Remove Initial Soluble P**: 10 g of air-dried soil (sieved to $<$ 2 mm) was suspended in 200 ml of deionized water and horizontally shaken for 60 minutes at 120 rpm. The suspension was centrifuged for 15 minutes at 4000 rpm, and the supernatant containing the readily soluble P was completely decanted. The P concentration of this discarded fraction was not quantified in this study, as the primary objective was to reset the soil solution to $P \approx 0$ to isolate the subsequent

solid-phase desorption kinetics. No soil was lost during decantation due to the highly compact nature of the pellet.

2. **Kinetic Extraction**: Because the remaining saturated soil pellet was highly compacted from centrifugation, achieving full resuspension required rigorous shaking, meaning the reactive surface area was not perfectly constant at the exact onset of the experiment. The pellet was rapidly resuspended in 200 ml of fresh deionized water to initiate the desorption process. The suspension was shaken continuously. To capture the nonlinear initial release phase, 2 ml subsamples were drawn at precisely eight time points: 2, 4, 10, 15, 20, 30, 45, and 60 minutes.
3. **Analysis**: Each subsample was immediately filtered through a 0.2 μm membrane filter to halt desorption, and the stoichiometric concentration of orthophosphate ($c(t)$) was determined colorimetrically using the malachite green method (Van Veldhoven & Mannaerts, 1987).

2.4.2 Temporal Stability of Amorphous Metal Oxides

Although the kinetic parameters (k) and geochemical structural variables (Fe_{ox}, Al_{ox}) were measured comprehensively in the year 2023, they were applied to the entire 26-year historical dataset. This is justified because oxalate-extractable iron and aluminum represent native, poorly-crystalline structural minerals rather than highly dynamic labile nutrient pools. Under constant arable management, these amorphous phases are widely recognized as pedologically stable over decadal timescales (Lookman et al., 1995; Schwertmann, 1964). Thus, the 2023 measurements serve as robust, invariant site-means to characterize the fundamental sorption capacity of each experimental field.

2.5 Statistical Analysis and Modeling

All data processing, mathematical modeling, and visualization were conducted using the R programming language (v. 4.3.1) (R Core Team, 2026). Primary packages included `nlme` (Pinheiro et al., 2026) for non-linear kinetic modeling, `lme4` (Bates et al., 2026) for linear mixed-effects modeling, and `deSolve` and `ReacTran` for partial differential equation solving.

2.5.1 1. Modeling of Desorption Kinetics and the Desorption Velocity (v_0)

To establish the foundational kinetic parameters, physical desorption curves were generated using a modified sequential extraction protocol based on Flossmann & Richter (1982). Soil samples were first pre-washed to discard readily soluble P, then resuspended in deionized water and continuously shaken. Subsamples were extracted across a high-resolution time series (2, 4, 10, 15, 20, 30, 45, and 60 minutes), and the orthophosphate concentration was determined colorimetrically using the malachite green method.

Deriving the Desorption Rate Constant (k): A non-linear mixed-effects model (`nlme`) was fitted to these high-resolution extraction curves. The desorption process was modeled as a hyperbolic, asymptotic saturation curve derived from the exact solution of the first-order rate equation:

$$c(t) = c_{max} \times (1 - e^{-k \times t'})$$

Where $c(t)$ is the desorbed P concentration at time t, c_{max} is the maximum size of the desorbable P pool (the asymptote), k is the first-order desorption rate constant, and t' is an adjusted time ($t_{min} + 3$ min) representing the empirically determined handling and resuspension time delay required to break apart the centrifuged pellet before true, uniform first-order desorption commenced. Sample-specific deviations were managed with the nested random effect structure `c_max + k ~ 1 | site / plot`, allowing both parameters to vary freely across sites and individual field plots.

Upscaling via Pedotransfer Function (PTF): Because this labor-intensive 60-minute kinetic extraction cannot be retroactively applied to the entire 30-year historical STYCS archive, a fundamental **Pedotransfer Function (PTF)** was constructed to predict the rate constant for missing soils and treatments. To ensure the upscaling process remained rigorously grounded in established mechanistic physics rather than arbitrary empirical correlations, the PTF was structured upon the Arrhenius equation—where reaction rates are driven by the competing activation energies of dominant sorption surfaces. Accordingly, a linear model (`lm`) was trained to predict $\ln(k)$ using explicitly pedogenetic predictor variables: the natural log of the ratio of amorphous aluminum to iron oxides ($\ln(Al_{ox}/Fe_{ox})$) and soil pH.

By anchoring the PTF to these inherently pedogenetic (treatment-independent) geochemical traits, we successfully extrapolated a predicted rate constant (k_{pred}) for every spatial and temporal plot in the dataset.

From these predicted parameters, a novel proxy for agronomic availability, the **Dynamic Desorption Velocity (v_0)**, was defined mathematically as the product of the predicted rate constant and the historically measured static concentration pool:

$$v_0 = k_{pred} \cdot P_{CO_2}$$

2.5.2 2. Thermodynamic $Q(I)$ Isotherms and Buffering Capacity

To deduce the Phosphorus Buffering Capacity (PBC), a thermodynamic Quantity-Intensity (Q/I) framework was constructed. The P_{AAE10} extraction was utilized as the standard mass Quantity (Q, mg P kg^{-1}), while the Davies-corrected a_{max} was utilized as the native Intensity (I, μM). It must be acknowledged that relying on P_{AAE10} as a proxy for absolute thermodynamic Quantity carries inherent mechanistic limitations. As demonstrated by isotopic exchange kinetics (IEK) studies, the extraction fluid's EDTA chelator actively dissolves non-exchangeable calcium phosphates—particularly in high-pH or high-CEC clays—leading to isotopic recovery rates as low as 25% (Demaria et al., 2005). Furthermore, in heavily buffered clay soils, the chelator physically saturates when overwhelmed by exchangeable background cations. Once saturated, the test becomes entirely "blind" to any further thermodynamic Q, artificially collapsing the measured ΔQ parameter.

The relationship between these variables across the varying fertilizer treatments at each site was parameterized using a log-linearized **Freundlich Isotherm** (Freundlich, 1906), which is universally recognized for describing ion sorption on heterogeneous soil surfaces:

$$Q = K_f \cdot I^{\frac{1}{n}}$$

The apparent Phosphorus Buffering Capacity (PBC) for each specific site and state was then calculated analytically as the first derivative of the fitted Freundlich isotherm with respect to Intensity:

$$\text{PBC} = \frac{dQ}{dI} = K_f \cdot \frac{1}{n} \cdot I^{(\frac{1}{n}-1)}$$

Predicting Quantity via Pedotransfer Functions (PTF) To evaluate the practical scalability of these thermodynamic concepts, empirical Quantity (P_{AAE10}) was modeled directly from Intensity using a log-linear mixed-effects model (`lme`):

$$\ln(Q) = \beta_0 + \beta_I \cdot \ln(I) + \sum(\beta_j \cdot Z_j) + \ln(I) \sum(\beta_{int,j} \cdot Z_j) + u_{site} + u_{plot|site}$$

- **The Covariates (Z_j):** Geochemical pedotransfer variables including soil fine texture (clay + silt), pH, C_{org}, and exchangeable background cations (Ca, Mg, K). These are mechanistically justified as they structurally determine the density and binding energy of the soil's sorption complexes (e.g., amorphous metal oxides and calcium carbonates).
- **Interaction Terms:** The covariates directly interact with Intensity because the *slope* of the Q/I isotherm (the buffer power) is fundamentally altered by these intrinsic geochemical properties.
- **Random Effects ($u_{year|site}$):** Nested temporal and spatial intercepts designed to rigorously absorb unmeasured macro-climatic shifts and temporal biological chaos (e.g. frost timing, pest pressure, radiation) across the different experimental years, preventing pseudo-replication.

2.5.3 3. Agronomic Normalization: Controlling Pedoclimatic Noise

To rigorously separate underlying P-limitation from the immense baseline yield variance caused by pedoclimatic shifts and crop genetics, all raw yield and P-uptake metrics were normalized to a biologically strict empirical ceiling. For every combination of site, year, and crop, the absolute maximum value observed across all fertilization treatments was assigned as 100% capacity.

$$Y_{rel} = \frac{Y_{plot}}{\max_{treatment}(Y)_{site,year,crop}}$$

Unlike using a specific fertilization label (e.g., the P166 treatment average), this method guarantees that the phenomenological Mitscherlich asymptote of 1.0 is strictly bounded. It ensures that the unconstrained biological ceiling for that specific environment is defined by the actual absolute maximum biomass it produced, structurally eliminating random negative constraints (e.g., localized compaction or pest damage) that might artificially depress a specific treatment average. This mathematical operation removes the variance in the absolute biological ceiling, strictly isolating the chemical P-limitation from the overarching temporal pedoclimatic noise.

2.5.4 4. Comparative Modeling of Agronomic Outcomes

To test the core hypothesis, biological response variables were modeled using non-linear mixed-effects models (`nlme`), recognizing that plant growth and nutrient uptake follow saturating, asymptotic biological curves rather than linear relationships.

1. The Mitscherlich Yield Model The agronomic yield response was modeled using the classical Mitscherlich equation. The explicit mathematical formulation replaces the static rate constant with a dynamic effective rate (c_{eff}):

$$Y_{rel} = 1 - \exp\left(-c_{eff} \cdot (\mathrm{STP} + E_{base})\right)$$

Where c_{eff} is dynamically modulated by a set of fixed pedoclimatic covariates (Z_i) and nested spatial random effects (u):

$$c_{eff} = (c_{base\_crop} + u_{site} + u_{plot|site}) \cdot \exp\left(\sum \beta_i \cdot Z_i\right)$$

- **The Diffusion Bottleneck (β_{invb}):** The inverse physical buffer power ($1/b$) acts as the primary transport penalty. Soils with high buffer power cannot replenish the root depletion zone fast enough via diffusion, functionally decreasing the agronomic efficiency of the measured STP.
- **Speciation and Solubility (β_{pH}):** Soil pH fundamentally controls the speciation of orthophosphate in solution and drives precipitation dynamics, altering thermodynamic stability.
- **Stoichiometric Co-Limitation ($\beta_N, \beta_K, \beta_{Mg}$):** Following Liebig's Law of the Minimum, the plant's ability to efficiently convert P into biomass is inherently capped if Nitrogen, Potassium, or Magnesium are limiting.
- **Pedoclimatic Noise ($\beta_{Temp}, \beta_{Prec}$):** Temperature and precipitation anomalies dictate physical soil moisture (governing diffusion transport) and the basic biological metabolic rate.

2. The Plant Uptake Model (Michaelis-Menten) Actual mass-flux of P into the plant was modeled using a fractional Michaelis-Menten biological limit:

$$Uptake_{rel} = \frac{U_{eff} \cdot \mathrm{STP}}{K_m + \mathrm{STP}}$$

Where the effective maximum uptake capacity (U_{eff}) expands upon the base physiological limit (U_{base}) via critical covariates:

$$U_{eff} = U_{base} + \beta_{temp} \cdot Z_{temp} + \beta_N \cdot Z_N + \beta_{v0} \cdot Z_{v0} + u_{site} + u_{year|site}$$

- **The Kinetics (β_{v0}):** The dynamic desorption velocity (v_0) directly increases maximal uptake capacity by continuously resupplying the root depletion zone.
- **Crop-Specific Affinity (K_m):** The half-saturation constant (K_m) is modeled as a fixed effect grouped by *crop species*, reflecting the differing physiological root architectures and transporter affinities of maize, wheat, etc.
- **Random Effects ($u_{year|site}$):** Nested `site/year` intercepts absorb the massive macro-climatic shifts (e.g., a regional drought year systemically lowering the overall U_{base}).

3. The Tissue Concentration Model (C_P) To expose the physiological dynamics of tissue concentrations, the apparent C_P was modeled phenomenologically. Rigorous mixed-effects testing revealed that the classical Piper-Steenbjerg carbon dilution penalty was statistically undetectable across pedoclimatic noise. Instead, the phenomena clearly dictate a simpler biological reality: A crop-specific baseline tissue concentration (C_{base}), followed by a linear **Luxury Accumulation** phase (S_{acc}):

$$C_P = C_{base} + S_{acc} \cdot \mathrm{STP}$$

- **Crop-Specific Baseline (C_{base}):** The fundamental, minimum tissue concentration required for the plant to physically exist, modeled as a fixed effect grouped by crop species.
- **Luxury Accumulation (S_{acc}):** Once basic metabolic needs are met, plants linearly accumulate "luxury" P in their tissues (primarily in the vegetative biomass). This slope (S_{acc}) is significantly driven by the soil's desorption velocity (v_0) and buffer power ($1/b$), which dictate how rapidly the soil can push excess phosphorus into the plant.
- **A Phenomenological P_{crit}:** Because field data rarely captures conditions severe enough to observe true Piper-Steenbjerg dilution, defining P_{crit} via abstract mathematical minima of C_P curves is flawed. Instead, we define P_{crit} phenomenologically: as the threshold where the plant has achieved its physiological baseline (C_{base}) *plus* a defined luxury reserve buffer (e.g., x%) strictly necessary to survive difficult environmental phases like pH swings or transient droughts.

Table 3: Comprehensive description of all response variables and covariates used in the thermodynamic and agronomic models, including their explicit preprocessing pipelines.

Abbreviation	Variable	Description	Processing and Filtering
Y_{rel}	Relative Yield	Plot yield normalized by the local year's maximum.	Normalized by the mean of the P166 treatment within the specific site × crop × year. Filtered for finite values.
$Uptake_{rel}$	Relative P Uptake	Plot P-uptake normalized by the local year's maximum.	Normalized by the local P166 maximum. Rows with 0 uptake dropped.
$\ln(P_{AAE10})$	Log Quantity Pool	Standard mass of extractable P (AAE10 method).	Natural log transformed to ensure strictly positive predictions and stabilize variance in mixed models.
P_{CO2}	Empirical Intensity	Water-extractable P (CO_2 saturated).	Raw measured values. Log-transformed for PTF training.
a_{CO2}	Thermodynamic Intensity	Thermodynamic activity of the water pool.	Calculated via Davies equation accounting for ionic strength. Log-transformed.
k	Desorption Rate	First-order rate constant of P desorption.	Missing historical plots were imputed using a pedogenetic PTF trained on log(Al/Fe) + pH. Standardized (z-score).
v_0	Desorption Velocity	Initial dynamic supply rate.	Calculated as product of predicted k and empirical P_{CO2}. Standardized (z-score).
PBC^{-1} ($1/b$)	Inverse Buffer Power	Inverse of the dynamic Q/I slope (dI/dQ).	Analytically derived from predicted Freundlich isotherms. Standardized (z-score).
Fe_{ox}, Al_{ox}	Amorphous Oxides	Amorphous Iron and Aluminum oxides.	Averaged into invariant site-means to remove management artifacts, log-transformed, and standardized.
Fine Texture	Clay + Silt	Sum of clay and silt fractions.	Log-transformed sum to linearize bounds, standardized. Missing values imputed with site-means.
pH	Soil pH (H_2O)	Soil pH measured in water.	Standardized directly.
C_{org}	Organic Carbon	Total organic carbon.	Log-transformed and standardized. Missing values imputed with site-means.

Abbreviation	Variable	Description	Processing and Filtering
Ca, Mg, K	Exchangeable Cations	AAE10-extractable background cations.	Missing values imputed using site medians (or global medians if site missing). Log-transformed and standardized.
$Temp_{Anom}$, $Prec_{Anom}$	Climate Anomalies	Juvenile phase temp/prec deviations.	Calculated as absolute deviation from the site's long-term historical mean. Standardized.
Fert N	Nitrogen Fertilizer	Total applied nitrogen fertilizer.	Standardized directly.

2.5.5 5. Mechanistic Modeling of the Rhizosphere (Barber-Cushman Modification)

To mechanistically validate the field-scale statistical findings and visualize the limitations of static buffering capacity, a modified radial transport model based on Barber-Cushman principles (Barber & Cushman, 1981) was constructed. Standard iterations of the Barber-Cushman model calculate radial diffusion using stoichiometric concentration (C_l) and a static buffer power ($b = \Delta Q/\Delta I$), effectively assuming instantaneous soil-solution equilibrium.

To reflect the kinetic bottlenecks identified in this study, the partial differential equation (PDE) for radial transport in cylindrical coordinates was modified. The driving gradient was kept as empirical concentration (c_l), and the static buffer power was replaced with a dynamic, first-order kinetic source term ($R_{desorption}$):

$$\frac{\partial c_l}{\partial t} = D_e \left(\frac{\partial^2 c_l}{\partial r^2} + \frac{1}{r}\frac{\partial c_l}{\partial r} \right) + \frac{r_0 v_0}{r}\frac{\partial c_l}{\partial r} + R_{desorption}$$

Where D_e is the effective diffusion coefficient, r is the radial distance, r_0 is the root radius, and v_0 is the water flux at the root surface. The kinetic source term resupplying the solution was defined using the experimentally derived parameters:

$$R_{desorption} = k \cdot (c_{max} - c_l)$$

Furthermore, the inner boundary condition at the root surface ($r = r_0$) was upgraded to reflect a dual-affinity Michaelis-Menten uptake system (Michaelis & Menten, 1913), accounting for both high-affinity (PHT1) and low-affinity (LATS) transporter kinetics. The binding kinetics of these transporters are governed by the fundamental thermodynamic driving force of the substrate: its chemical activity (a), not its total stoichiometric concentration (c) (Atkins & Paula, 2011; Minton, 2001). The boundary condition was defined as:

$$J_r = \frac{V_{max1} \cdot (c_l - c_{min})}{K_{m1} + (c_l - c_{min})} + \frac{V_{max2} \cdot (c_l - c_{min})}{K_{m2} + (c_l - c_{min})}$$

The modified PDEs were solved numerically in R utilizing the `ReacTran` and `deSolve` packages to generate dynamic spatial profiles of the P-depletion zone over time.

2.5.6 6. Derivation of Critical Thresholds (P_{crit})

To translate the phenomenological models into actionable agronomic metrics, the critical Soil Test Phosphorus (P_{crit}) threshold was analytically derived from all three modeling frameworks. Rather than relying on arbitrary inflection points, each derivation represents a distinct physiological or agronomic paradigm:

1. The Relative Yield Paradigm (The Production Target) Agronomically, P_{crit} is classically defined as the STP required to achieve a specific operational threshold, commonly set as an $x\%$ relative yield target (q_Y, e.g., 0.95). By setting the relative yield target to q_Y in the full Mitscherlich equation ($Y_{rel} = 1 - \exp(-c_{eff} \cdot (\text{STP} + E_{base}))$), the required soil concentration is extracted:

$$P_{crit\_Y} = \frac{-\ln(1 - q_Y)}{c_{eff}} - E_{base}$$

Where the effective rate constant incorporates the pedoclimatic penalties: $c_{eff} = c_{base} \cdot \exp(\beta_{kin} \cdot \text{Kinetics} + \dots)$.

2. The Tissue Concentration Paradigm (The Physiological Buffer) Because true Piper-Steenbjerg minima were statistically absent, the tissue concentration critical limit was redefined as the threshold necessary to guarantee the plant's fundamental metabolic baseline (C_{base}) *plus* a luxury reserve (γ, representing an $x\%$ safety buffer). Solving the linear mixed-effects model ($C_P = C_{base} + S_{acc} \cdot \text{STP}$) for the target $C_{base}(1 + \gamma)$ yields:

$$P_{crit\_C} = \frac{\gamma \cdot C_{base}}{S_{base} + \beta_{v0} \cdot v_0 + \beta_{invb} \cdot \frac{1}{b}}$$

This derivation explicitly demonstrates that soils with a faster desorption velocity (v_0) or stronger buffer power ($1/b$) create a steeper accumulation slope (S_{acc}), inherently reducing the P_{crit} required in the soil to safely buffer the crop against stress.

3. The Relative Uptake Paradigm (The Mass-Transfer Limit) While yield integrates the entire season, instantaneous phosphorus uptake is strictly governed by Michaelis-Menten kinetics at the root surface. Defining a critical threshold as the STP required to reach an $x\%$ relative uptake target (q_{up}) yields a direct relationship with the effective half-saturation constant (K_{eff}):

$$q_{up} = \frac{P_{crit\_up}}{K_{eff} + P_{crit\_up}} \implies P_{crit\_up} = \left(\frac{q_{up}}{1 - q_{up}}\right) K_{eff}$$

Where K_{eff} structurally expands upon the base crop affinity (K_m) via the same physical kinetic bottlenecks identified in the field data.

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4 Appendix: Thermodynamic Correction of P_{CO_2} Extractions

4.1 A.1. Intrinsic Ionic Strength of the P_{CO_2} Extractant

Standard static soil test methods rely on the operational extraction of phosphorus. Strong extractants (e.g., Olsen P, P-AAE10) possess inherently high ionic strengths that dominate the soil matrix, complicating absolute thermodynamic comparisons without complex modeling. In contrast, the P_{CO_2} method utilizes a very weak extractant: water saturated with carbon dioxide at atmospheric pressure (1 atm) and 25 °C.

To demonstrate that the P_{CO_2} extractant adopts the native ionic strength of the soil, we calculate the intrinsic ionic strength (I) of the pure extractant.

According to Henry's Law, the concentration of dissolved CO_2 (denoted as $H_2CO_3^*$) is:

$$[H_2CO_3^*] = K_H \times P_{CO_2}$$

Given Henry's constant ($K_H \approx 0.034$ M atm^{-1}) and $P_{CO_2} = 1$ atm, the molarity of dissolved CO_2 is approximately 0.034 M.

The primary ionic species generated arise from the first dissociation of carbonic acid:

$$H_2CO_3^* \rightleftharpoons H^+ + HCO_3^-$$

Using the acid dissociation constant ($K_{a1} = 4.46 \times 10^{-7}$ at 25 °C):

$$K_{a1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3^*]}$$

Because $[H^+] \approx [HCO_3^-]$, we can solve for the bicarbonate concentration:

$$[HCO_3^-] = \sqrt{4.46 \times 10^{-7} \times 0.034} \approx 1.23 \times 10^{-4} \text{ M}$$

The intrinsic ionic strength (I) of the extractant is defined as:

$$I = \frac{1}{2}\sum(C_i z_i^2)$$

$$I = \frac{1}{2}\left((1.23 \times 10^{-4} \times (+1)^2) + (1.23 \times 10^{-4} \times (-1)^2)\right) \approx 0.00012 \text{ M}$$

This intrinsic ionic strength (≈ 0.12 mM) is negligible compared to native agricultural soil solutions (which typically range from 10 to 50 mM). Therefore, the final ionic strength of the P_{CO_2} extract is primarily dictated by the dissolved soluble salts of the specific soil being analyzed.

4.2 A.2. Justification for the Davies Equation

Because the P_{CO_2} extract adopts the native ionic composition of the soil, reading the orthophosphate concentrations against an $I = 0$ (distilled water) calibration curve introduces systematic error. The fundamental thermodynamic driving force for P-availability is the chemical **activity**, not the stoichiometric concentration.

To correct for this, an activity coefficient (γ) must be applied. While the Pitzer equations are the standard method for high-ionic-strength solutions ($I > 0.5$ M, such as the P-AAE10 extractant), the ionic strength of native agricultural soil solutions is typically much lower ($I < 0.1$ M).

In this low-to-moderate ionic strength regime, the **Davies equation** (an empirical extension of the Debye-Hückel theory) is highly accurate and does not require complex, ion-specific interaction parameters.

The Davies equation calculates the activity coefficient (γ) as follows:

$$\log_{10}(\gamma) = -Az^2\left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3I\right)$$

Where: * A is a temperature-dependent constant (≈ 0.509 L$^{0.5}$mol$^{-0.5}$ for water at 25 °C). * z is the charge of the ion in question (for orthophosphate, z depends on pH, predominantly -1 for $H_2PO_4^-$ and -2 for HPO_4^{2-}). * I is the total ionic strength of the extracted solution, calculated from the measured concentrations of major cations (Ca^{2+}, Mg^{2+}, K^+).

4.3 A.3. Implementation of Thermodynamic Correction

To assess the theoretical P-Intensity (I) and formulate the Phosphorus Buffering Capacity (PBC), the operationally defined P_{CO_2} concentrations are converted to thermodynamic activities using the following sequential steps:

1. **Determine Total Ionic Strength (I):** Using the measured cation concentrations from the extracts.
$$I = \frac{1}{2}\left([K^+] + 4[Ca^{2+}] + 4[Mg^{2+}] + ...\right)$$
2. **Calculate the Activity Coefficient (γ):** Applying the Davies equation based on the sample's specific I.
3. **Compute Chemical Activity:**

$$\text{Activity}(P) = [P_{CO_2}] \times \gamma$$

By standardizing the measurements to their corrected thermodynamic activity, we remove the analytical bias introduced by varying soil salinity and pedoclimatic history, allowing for an absolute comparison of P-Intensity across all STYCS experimental sites. ## A.4 Supplementary Materials and Reproducibility

This Master's thesis was produced using a fully reproducible workflow with the **Quarto** publishing system. All data, R scripts, and analytical notebooks used to generate the figures, tables, and results presented in this work are openly available.

The complete project can be cloned from the author's GitHub repository, which contains the raw data, the R code for the kinetic and statistical models, and the Quarto source files.

GitHub Repository URL: [https://github.com/ptochosgeorgos/Article-P-desorption-kinetics]

A rendered version of the full project, including the analytical notebooks that document the development process, is also available as a GitHub Pages website at the following URL:

GitHub Pages Site URL: [https://ptochosgeorgos.github.io/Article-P-desorption-kinetics/]

This approach ensures full transparency and allows for the complete replication of the findings presented in this thesis.